instead have been conceived ‘as a supplement to rather than a substitute for the market economy’—launched the Swatantra Party, whose founding statement called for the ‘decentralised distribution of industry’, promotion of ‘competitive enterprise’ and rejection of ‘so-called socialism’ and ‘“Statism”’. [53] For various other reasons, India was also unable to take full advantage of the technological access and benefits it had accrued diplomatically. In the early 1960s, a high-profile review under Nobel Prize-winning British scientist Patrick Blackett presented a bleak picture of the state of scientific progress in India. In his assessment, the administration of India’s research laboratories, brain drain, and lack of applied research were contributing to scientific and technological failure. [54]

Atomic Infancy: The Nuclear Programme, 1947–1971

One area of scientific and technical progress that assumed immediate interest upon independence involved nuclear technologies, which held significant potential both for electricity generation and military applications. The origins of India’s atomic energy programme can be traced to the initial efforts of Homi Bhabha. In 1944, Cambridge-educated Bhabha convinced the Tata Trust to fund the Tata Institute of Fundamental Research, which opened the next year and marked the beginnings of established nuclear research in India. Shortly after Independence in 1948, Nehru created the Atomic Energy Commission with Bhabha at its helm. Bhabha understood that an Indian nuclear energy programme was impossible without foreign support in the form of reactor designs, financing, and fuel. This required India to enter into commitments to use foreign-supplied material only for peaceful purposes.

From the 1950s onwards, India benefited from US, Canadian, British, and French nuclear assistance and interest from Soviet nuclear scientists. In 1955, the Apsara research reactor began construction and was supplied with uranium from the United Kingdom. The same year, Canada agreed to help build a larger reactor, eventually called the Canadian–Indian Reactor, US (CIRUS), given that the United States supplied heavy water, containing